



Certiguard: a fine-tuned LLM for mastering security certification exams

Chun-Ming Lai¹ · Lien-Jung Chang¹ · Chun-Chieh Chang¹ · Yi-Chao Wu²

Received: 9 September 2025 / Accepted: 3 August 2026
© The Author(s) 2026

Abstract

Cybersecurity certification exams have become an important pathway for developing professional competence, yet preparation for exams such as CISA, CRISC, CEH, and CCNA often relies on fragmented materials, costly training programs, and limited individualized feedback. This study introduces CertiGuard, a resource-efficient cybersecurity training framework built on LLaMA-2-7B. The model is adapted through a two-stage process: continued pre-training on a curated cybersecurity corpus, followed by supervised fine-tuning with certification-aligned instructions. To reduce the computational cost of domain adaptation, the framework applies QLoRA-based fine-tuning and organizes model use through beginner, intermediate, and advanced proficiency tiers. Experiments were conducted on four certification-oriented question-answering datasets covering auditing, risk and control, ethical hacking, and networking domains. The base Certibase model achieved an average accuracy of 54.1%, while the best structured prompt improved the average accuracy to 72.0%. Further evaluation of reasoning strategies showed that self-consistency Chain-of-Thought achieved the strongest overall performance, reaching an average accuracy of 86.5% across the four exam domains. The results suggest that domain adaptation, structured prompting, and reasoning-oriented inference can substantially improve LLM performance on cybersecurity certification questions. At the same time, the study highlights practical limitations related to small evaluation sets, dataset provenance, copyright restrictions, and the need for broader human-centered evaluation. Overall, CertiGuard provides a lightweight and scalable approach for certification-oriented cybersecurity training, while offering a basis for future adaptive learning systems and reproducible evaluation benchmarks.

Keywords Large language models · Cybersecurity certification exams · Fine-tuning · Chain-of-thought reasoning

1 Introduction

Cybersecurity has become a core requirement for individuals, organizations, and governments as digital services increasingly depend on secure networks, software, cloud platforms,

and governance processes. At the same time, the cybersecurity workforce shortage continues to make training and certification preparation difficult to scale. Professional certifications such as Certified Information Systems Auditor (CISA), Certified in Risk and Information Systems Control (CRISC), Certified Ethical Hacker (CEH), and Cisco Certified Network Associate (CCNA) are widely used to structure learning outcomes and validate professional competency. These certifications cover complementary domains: auditing and governance, risk and control, ethical hacking and penetration testing, and networking fundamentals. Preparing for these examinations is therefore not merely a question-answering task; it requires learners to connect conceptual knowledge, scenario-based judgment, and domain-specific terminology.

Traditional certification preparation typically relies on textbooks, instructor-led bootcamps, commercial question banks, and static practice tests. These resources are useful, but they also have limitations: they can be expensive,

✉ Yi-Chao Wu
alanwu@ntut.edu.tw

Chun-Ming Lai
cmlai@thu.edu.tw

Lien-Jung Chang
s09350148@thu.edu.tw

Chun-Chieh Chang
g11350033@thu.edu.tw

¹ Department of Computer Science, Tunghai University, No.1727, Section 4, Taiwan Boulevard, Xitun District, Taichung City 407224, Taiwan

² Department of Electronic Engineering, National Taipei University of Technology, No. 1, Section 3 Zhongxiao East Road, Da'an District, Taipei City 106335, Taiwan

difficult to update, and insufficiently responsive to learners with different starting points. In parallel, large language models (LLMs) have shown strong capabilities in natural language understanding, explanation generation, and multi-step reasoning [1–3]. In educational settings, LLMs and ChatGPT-style systems are increasingly studied as tools for feedback, tutoring, assessment support, and activity-based learning [4–7]. These developments motivate the question of whether a domain-adapted LLM can support cybersecurity certification learning in a scalable and resource-efficient manner.

The problem addressed in this study is the development and evaluation of a lightweight, certification-oriented cybersecurity language model that can answer exam-style questions and provide explanatory feedback while remaining feasible to fine-tune under limited computational resources. We focus on four research questions:

- How can a general LLM be adapted to cybersecurity certification knowledge using a resource-efficient fine-tuning pipeline?
- How do certification-aligned prompts influence model performance across managerial and technical security domains?
- How do different reasoning strategies, including Chain-of-Thought variants, affect exam-style question answering?
- What are the limitations, reproducibility constraints, and ethical considerations of using certification-style materials for model training and evaluation?

To answer these questions, we propose *CertiGuard*, a cybersecurity training framework based on LLaMA-2-7B and QLoRA [8, 9]. The model is developed through continued domain adaptation on cybersecurity materials and supervised fine-tuning with certification-aligned instructions. We evaluate prompt designs and reasoning strategies across CISA, CRISC, CEH, and CCNA exam-style datasets. The study addresses three important aspects of the proposed framework and its evaluation. First, the deployment strategy is described as *proficiency-tiered* rather than fully adaptive personalization. Second, the data section distinguishes between publicly available certification outlines, third-party copyrighted study materials, and non-redistributable exam-style question text. Third, the results are interpreted with statistical uncertainty because the evaluation sets are small.

The contributions of this paper are as follows:

- We present a two-stage, parameter-efficient fine-tuning pipeline for cybersecurity certification learning.
- We compare structured prompt designs and multiple reasoning strategies across four certification domains.

- We provide a domain coverage map linking cybersecurity certification areas to governance, risk, compliance, network security, cryptography, ethical hacking, cloud/application security, and operational security.
- We add reproducibility safeguards, prompt templates, uncertainty analysis, and a detailed discussion of data provenance, copyright constraints, limitations, and ethical risks.

The remainder of this paper is organized as follows. Section 2 reviews related work on LLMs, education, professional certification assessment, cybersecurity training, and limitations of current approaches. Section 3 describes the proposed methodology, including domain coverage, data corpus construction, fine-tuning, prompt designs, and leakage prevention. Section 4 presents the experimental configuration and evaluation protocol. Section 5 reports the results and qualitative analysis. Section 6 discusses explainability, ethics, deployment scenarios, limitations, and future work, and Sect. 7 concludes the paper.

2 Related work

Large Language Models (LLMs) have attracted considerable attention across education, cybersecurity, medicine, and materials science. Table 1 compares representative domain-adapted LLM studies and highlights whether they use continuous training, domain-specific fine-tuning, and reasoning prompts.

2.1 Evolution of large language models

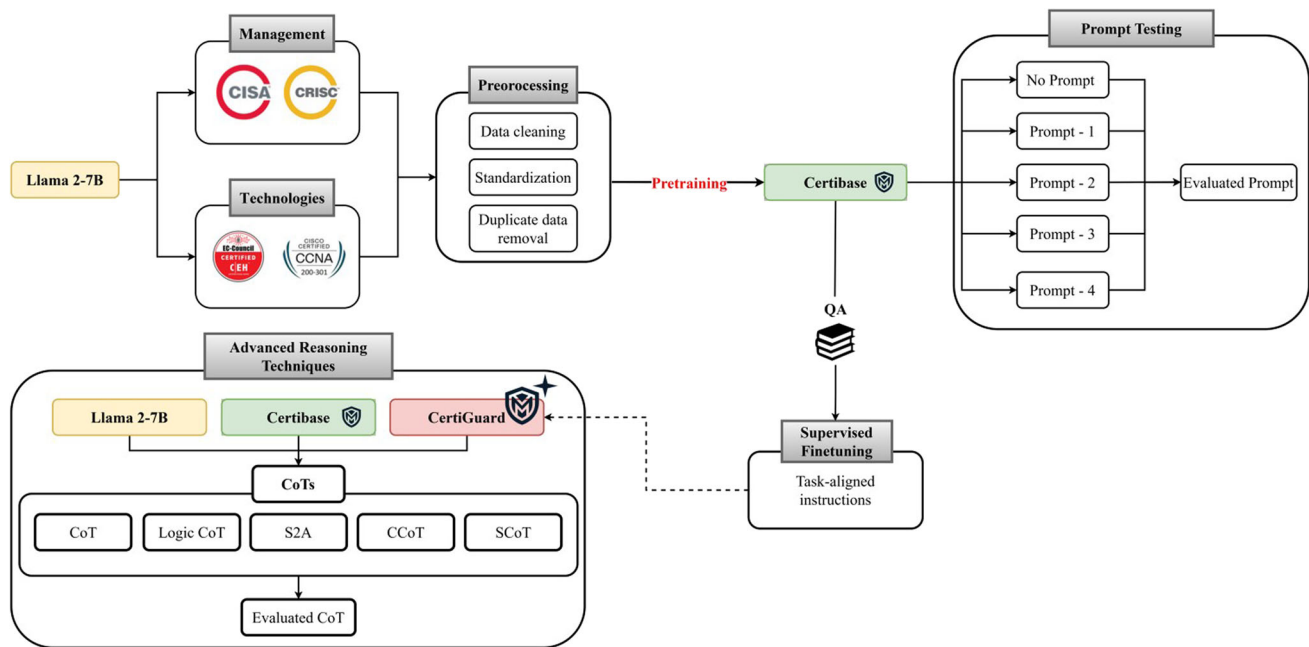
The development of LLMs can be traced from sequence models and attention mechanisms to Transformer-based pre-training. The Transformer architecture introduced self-attention as a scalable mechanism for learning contextual representations [1]. Later, large-scale generative pre-training demonstrated that increasing model and data scale can produce broad language capabilities [2]. Instruction tuning and human feedback further improved the usability and alignment of language models for interactive tasks [3]. More recently, parameter-efficient fine-tuning methods such as LoRA and QLoRA have made it possible to adapt large models to specialized domains with substantially reduced hardware requirements [8, 9]. These developments provide the technical foundation for applying LLMs to domain-specific cybersecurity education.

2.2 AI and LLMs in education and assessment

LLMs are increasingly used in educational settings to generate explanations, provide formative feedback, support

Table 1 Comparison of large language models across different domains

Paper	Domain	Cont. train.	Domain FT	Base model	#CoT
[10]	Medical	Yes	Yes	LLaMA2	None
[11]	Medical	No	No	LLaMA	One
[12]	Materials science	No	Yes	LLaMA2-7B	None
Ours	Cyber security	Yes	Yes	LLaMA2-7B	Multiple

**Fig. 1** System pipeline

tutoring, and assist assessment design. Kasneci et al. discuss opportunities and challenges of ChatGPT-like systems in education, including personalization, feedback quality, and academic integrity [4]. Activity-based learning combined with ChatGPT has also been studied as a way to improve student engagement and performance [5]. Broader reviews emphasize that LLMs can support learning environments but must address hallucination, bias, privacy, and overreliance risks [6, 7]. Educational theory is also relevant: immersive and experiential learning research, including virtual-reality laboratories, shows that learning tools should be aligned with cognitive load, feedback, learner agency, and situated practice [13]. These findings motivate the design of CertiGuard as an explanatory support tool rather than as a replacement for instructors or validated certification training.

2.3 Existing tools for certification exam preparation

Professional certification preparation commonly relies on official exam blueprints, review manuals, commercial courses, flashcards, practice exams, and question banks. These tools are valuable because they align learning objectives with exam domains, but they are often static and may

provide limited explanation for why an option is correct or incorrect. Moreover, commercial certification tools are typically closed systems, making it difficult to compare models under a standardized research protocol. In cybersecurity, this challenge is amplified by the fact that exam items are usually protected as intellectual property and may be subject to non-disclosure agreements. Therefore, research systems such as CertiGuard must balance practical relevance with responsible data use and reproducibility.

2.4 LLMs in cybersecurity and professional exams

LLMs have been explored for cybersecurity tasks such as threat intelligence, vulnerability analysis, phishing detection, capture-the-flag exercises, and certification question answering. SecureBERT adapts language modeling to cybersecurity text analytics and demonstrates the value of domain-specific representations [14]. Crimson shows that retrieval-aware training can improve strategic reasoning for cybersecurity defense tasks [15]. Tann et al. evaluate LLMs on capture-the-flag and certification-style questions and report that LLMs perform better on factual questions than on conceptual or multi-step tasks [16]. These studies suggest that cyberse-

curity LLMs require not only domain vocabulary but also structured reasoning support.

2.5 Limitations of current approaches

Current approaches face several limitations. First, many LLM evaluations use small or non-standardized test sets, which makes reported gains difficult to compare. Second, certification-style datasets may include copyrighted or non-redistributable materials, limiting reproducibility. Third, models may memorize training examples or produce overconfident explanations that appear plausible but are wrong. Fourth, static question-answer systems provide limited learner modeling and cannot fully replace adaptive tutoring systems. These limitations directly shape the design and evaluation of CertiGuard: we frame it as a proficiency-tiered support tool, report statistical uncertainty, clarify data restrictions, and discuss ethical and trustworthiness issues.

3 Methodology

This section describes the design and implementation of the proposed cybersecurity training framework, *CertiGuard*. As shown in Fig. 1, the framework follows a multi-stage pipeline consisting of data collection and preprocessing, continued pre-training, supervised fine-tuning with task-aligned instructions, and proficiency-based deployment. We also include a prompt testing module to examine how different prompt designs and reasoning strategies affect model performance [17].

3.1 System overview

The system architecture of CertiGuard is presented in Fig. 1. The framework contains two main flows: model construction and prompt-based evaluation. In the model construction flow, cybersecurity topics are collected from four widely recognized certification domains, including CISA, CRISC, CEH, and CCNA. These certifications were selected because they cover both managerial and technical aspects of cybersecurity. CISA and CRISC focus more on auditing, governance, risk management, and control, whereas CEH and CCNA emphasize ethical hacking, network security, and networking fundamentals.

After topic selection, the collected materials are processed through data cleaning, standardization, and duplicate removal. This preprocessing step removes irrelevant or repeated content, normalizes terminology across different certification domains, and improves the consistency of the training corpus. The resulting corpus is then used for continued pre-training based on LLaMA-2-7B. This stage produces *Certibase*, a domain-adapted foundational model that is more

familiar with cybersecurity terminology, attack and defense concepts, governance practices, and networking knowledge.

The pre-trained *Certibase* model is then refined through supervised fine-tuning with task-aligned instructions. These instructions are designed according to the knowledge requirements of each certification domain, enabling the model to answer certification-style questions and provide explanations that are aligned with the corresponding cybersecurity context. In this stage, QLoRA is applied to reduce memory consumption during fine-tuning while preserving the benefits of domain adaptation. The resulting model, *CertiGuard*, is therefore designed not only to select answers for exam-style questions, but also to provide supporting explanations for learners.

The prompt testing flow is shown on the right side of Fig. 1. This module evaluates different prompt designs, including the no-prompt baseline and Prompt-1 to Prompt-4, to compare how instruction structure affects model performance. In addition, several reasoning strategies are examined, including Chain-of-Thought, Logic CoT, System 2 Attention, Contrastive CoT, and Self-consistent CoT [18–21]. This design allows us to evaluate the effects of both domain-specific fine-tuning and reasoning-oriented prompting on cybersecurity certification question answering.

Finally, CertiGuard is organized into beginner, intermediate, and advanced usage tiers. These tiers are intended to adjust the depth of explanation and the level of guidance, rather than to represent a fully adaptive personalization system. Beginner-level responses emphasize definitions and basic reasoning, while intermediate and advanced responses provide more detailed analysis, practical interpretation, and certification-oriented justification.

3.2 Instruction design

Table 2 summarizes the instructions developed for each certification dataset (CISA, CRISC, CEH, and CCNA). These instructions specify the context and expectations for answering multiple-choice questions, ensuring that the model aligns with the knowledge and standards outlined in each exam's official materials. The design of these instructions is guided by effective prompt engineering principles [17].

3.3 Security certification domain coverage

To address the representativeness concern raised by small certification-specific datasets, we explicitly map the certification sources to cybersecurity knowledge areas. Table 3 summarizes the domain coverage considered in this study. The goal is not to reproduce proprietary certification exams, but to align the training and evaluation tasks with broad competency areas commonly represented in official exam outlines and study materials.

Table 2 Instruction design for multiple cybersecurity certifications

Dataset	Instruction
<i>CISA</i>	You are a Certified Information Systems Auditor (CISA) candidate preparing for the CISA exam. You need to demonstrate your understanding of information systems auditing, control, and security. Show your ability to apply knowledge essential for auditing information systems, including governance, management, acquisition, development, operations, and protection of information assets. For the following multiple-choice question, select one correct answer from A to E. Base your answer on the current standards, best practices, and guidelines referenced in ISACA's CISA Review Manual
<i>CRISC</i>	You are a candidate for the Certified in Risk and Information Systems Control (CRISC) exam. You need to demonstrate your understanding of enterprise IT risk management, including identification, assessment, response, and monitoring. Show your ability to apply knowledge essential for designing and implementing information system controls to mitigate risk. For the following multiple-choice question, select one correct answer from A to E. Base your answer on the current standards, best practices, and guidelines referenced in ISACA's CRISC Review Manual
<i>CEH</i>	You are a candidate for the Certified Ethical Hacker (CEH) exam. You need to demonstrate your understanding of ethical hacking principles, including reconnaissance, scanning, enumeration, exploitation, and post-exploitation phases. Show your ability to apply knowledge essential for identifying vulnerabilities and securing information systems. For the following multiple-choice question, select one correct answer from A to E. Base your answer on the current standards, best practices, and guidelines referenced in the EC-Council's CEH curriculum
<i>CCNA</i>	You are a candidate for the CCNA 200-301 Implementing and Administering Cisco Solutions exam. You need to demonstrate your understanding of networking fundamentals, network access, IP connectivity, IP services, security fundamentals, automation, and programmability. Show your ability to apply the knowledge essential for implementing and administering Cisco network solutions. For the following multiple-choice question, select one correct answer from A to E. Base your answer on the current standards, best practices, and guidelines referenced in Cisco's CCNA study materials

3.4 Data corpus, provenance, and compliance

Table 4 provides an overview of the curated cybersecurity corpus used for domain adaptation. The corpus covers CISA, CRISC, CEH, and CCNA topics and includes both managerial and technical cybersecurity content. The pre-training corpus is designed to expose the model to terminology and concepts such as risk management, auditing, network security, ethical hacking, secure protocols, and compliance.

Table 3 Security certification domain coverage considered in CertiGuard

Domain	Representative topics	Primary sources
Network security	Network fundamentals, routing, switching, access control, IP services, secure connectivity	CCNA, CEH
Cryptography and PKI	Encryption, hashing, certificates, key management, secure protocols	CEH, CCNA
Ethical hacking and penetration testing	Reconnaissance, scanning, enumeration, exploitation, post-exploitation, vulnerability assessment	CEH
Cloud and application security	Web vulnerabilities, secure configuration, cloud security concepts, application risk	CEH, CCNA
Governance, risk, and compliance	IT governance, risk identification, risk response, control design, compliance requirements	CISA, CRISC
Audit and assurance	Information systems auditing, acquisition, development, operation, protection of information assets	CISA
Operational security	Monitoring, incident response concepts, security operations, defensive controls	CEH, CCNA, CISA

Table 4 Dataset statistics for training and validation

Dataset	Number of samples		Number of tokens	
	Train	Validation	Train	Validation
CISA	769	38 (5%)	321K	2.7K
CRISC	953	47 (5%)	384K	2.8K
CEH	709	35 (5%)	287K	2.9K
CCNA	763	38 (5%)	330K	2.6K
Total	3,194	158	1.33M	11K

We distinguish three categories of sources: (i) publicly available certification outlines and domain descriptions, (ii) publicly accessible study materials and educational explanations, and (iii) third-party exam-style question-answer materials. The first category is used to define domain coverage and learning objectives. The second category supports domain adaptation. The third category is used only for closed research evaluation and is not redistributed because certification exam items and practice questions may be protected by copyright, licensing terms, or exam security policies. To respect these restrictions, this study does not publish raw third-party question text and instead reports corpus statistics, prompt templates, evaluation procedures, and metadata-level descriptions.

Table 5 Training and testing distribution for certification datasets

Dataset	# Train samples	# Test samples (%)	Format
CISA	255	56 (21%)	Q + A
CRISC	258	52 (20%)	Q + A
CEH	257	53 (20%)	Q + A
CCNA	254	54 (21%)	Q + A

For supervised fine-tuning and evaluation, we used certification-aligned question-answer pairs summarized in Table 5. Because the test sets contain only 52–56 items per certification, all reported performance results should be interpreted as exploratory rather than definitive evidence of exam readiness.

3.5 Data filtering and leakage prevention

To reduce overfitting and potential leakage, the data preparation pipeline includes normalization, exact duplicate removal, and near-duplicate screening. Items are normalized by lowercasing, removing excessive whitespace, and stripping source-specific formatting. Exact duplicates are removed using text hashes. Near duplicates are screened by comparing normalized question stems and option strings; items with high lexical overlap or highly similar wording between training and testing partitions are flagged for manual review. This step is particularly important because certification preparation materials may contain similar scenario structures across sources. We also separate the discussion of domain coverage from the reproduction of proprietary question text, which helps ensure that the evaluation protocol remains transparent without redistributing restricted materials.

3.6 Pre-training

In the pre-training phase, *Certibase* is trained on the aggregated corpus using a Transformer-based architecture (LLaMA-2-7B). The objective is to enable the model to acquire a broad understanding of cybersecurity semantics, including risk assessment methodologies, network security protocols, and ethical hacking principles. During this stage, we also perform prompt testing on the textual data to assess the initial impact of various prompt designs (e.g., No Prompt, Prompt-1, Prompt-2).

3.7 Supervised fine-tuning

Once the foundational model is established, we refine it through supervised fine-tuning using task-aligned instructions, thereby evolving *Certibase* into *CertiGuard*. As illus-

trated in Fig. 1, *CertiGuard* integrates advanced reasoning techniques such as Chain-of-Thought (CoT) prompting [18], System 2 Attention [21], and Contrastive CoT [19]. Additionally, the use of QLoRA for parameter-efficient fine-tuning is implemented following the methodologies described in [8, 9]. In this stage, we employ question-answer pairs from the datasets summarized in Tables 4 and 5, combined with certification-specific instructions (see Table 2), to guide the model toward producing more accurate and detailed responses.

3.8 Proficiency-tiered cybersecurity language models

To accommodate learners with different backgrounds, we implement a proficiency-tiered deployment strategy for *CertiGuard*, as illustrated in Fig. 2. In contrast to a fully adaptive tutoring system, this strategy does not continuously infer a learner model or dynamically personalize all content. Instead, it provides three static but pedagogically differentiated configurations:

- *Beginner Level*: Focuses on foundational terminology, basic security concepts, and direct explanations.
- *Intermediate Level*: Combines conceptual explanation with practical scenario interpretation for learners with moderate cybersecurity proficiency.
- *Advanced Level*: Emphasizes complex, scenario-based reasoning and detailed justification for experienced professionals.

This distinction is important because the present system provides proficiency-based scaffolding rather than full adaptive personalization. Future versions may integrate learner modeling, longitudinal performance tracking, and adaptive recommendation mechanisms to support more interactive personalization.

3.9 Prompt testing

As depicted on the right side of Fig. 1, we evaluate multiple prompt designs (No Prompt, Prompt-1, Prompt-2, etc.) to assess their effectiveness in eliciting high-quality responses from *CertiGuard*. This module tests different prompting styles on both textual data and exam-style questions, examining metrics such as accuracy, coherence, and the depth of reasoning [17].

3.10 Summary

Overall, the proposed methodology emphasizes a multi-stage pipeline—from data collection to proficiency-based deployment—designed to optimize cybersecurity training. By



Fig. 2 Proficiency-based prompts for different levels

integrating domain-specific instructions, advanced reasoning techniques, and flexible prompt strategies, *CertiGuard* aims to deliver scalable, proficiency-tiered learning experiences that address the diverse needs of modern cybersecurity education.

4 Experiment

This section describes the experimental environment and the detailed training configuration used to implement the proposed cybersecurity training framework, *CertiGuard*. The implementation focuses on optimizing the fine-tuning process of the base model (LLaMA-2-7B) and effectively integrating advanced reasoning techniques [8, 9].

4.1 Experimental environment

The fine-tuning of the LLaMA-2-7B model was conducted on the Google Colab platform, which provided access to an NVIDIA A100 GPU equipped with 40GB of HBM2

memory. This high-performance hardware enabled efficient training of large-scale models while managing memory constraints. Key software tools and libraries utilized in the implementation include:

- *Transformers* library for model loading and tokenization.
- *PEFT* (Parameter Efficient Fine-Tuning) with LoRA configurations to reduce the number of trainable parameters [9].
- *BitsAndBytes* for implementing 4-bit quantization (NF4), further reducing VRAM usage [8].
- *TRL* (Transformers Reinforcement Learning) toolkit for managing the fine-tuning process.

4.2 Training configuration

The training configuration was carefully selected to balance efficiency and performance. Key hyperparameters and techniques include:

- *Batch Size*: 4 samples per GPU.

Table 6 Key training configuration

Parameter	Setting
Base Model	LLaMA-2-7B
GPU	NVIDIA A100 (40GB HBM2)
Batch Size	4 per GPU
Learning Rate	2×10^{-4} (cosine schedule)
Precision	FP16 Mixed Precision
Quantization	4-bit (NF4)
Gradient Checkpointing	Enabled
Optimizer	Paged AdamW (32-bit)

- **Learning Rate:** Set to 2×10^{-4} with a cosine learning rate schedule to ensure stable convergence.
- **Precision:** Mixed precision training was employed using FP16; BF16 was not enabled due to hardware compatibility.
- **Quantization:** 4-bit quantization (NF4) was applied via BitsAndBytes to significantly reduce memory usage [8].
- **Gradient Checkpointing:** Enabled to further optimize memory consumption during training [8].

Table 6 summarizes the key components of the training configuration.

4.3 Implementation workflow

The overall implementation workflow is depicted in Fig. 1 (refer to Sect. 3 for the system pipeline). The process can be summarized as follows:

1. **Data Preparation:** A high-quality cybersecurity corpus is curated from certification materials and undergoes thorough preprocessing.
2. **Pre-training:** The foundational model (*Certibase*) is pre-trained on the aggregated corpus using unsupervised learning.
3. **Supervised Fine-Tuning:** Task-specific fine-tuning is performed with certification-aligned instructions, integrating advanced reasoning techniques such as Chain-of-Thought prompting [18] and System 2 Attention [21].
4. **Prompt Testing:** Multiple prompt designs are evaluated on both textual data and exam-style questions to assess performance.

This implementation framework is designed to ensure that *CertiGuard* achieves both high accuracy and scalability while efficiently managing computational resources [8, 9].

Table 7 Aggregate prompt performance with 95% Wilson confidence intervals ($N = 215$)

Prompt setting	Average passing rate	95% CI
No prompt	51.5%	[45.0%, 58.2%]
Prompt-1	64.1%	[57.5%, 70.3%]
Prompt-2	63.6%	[57.0%, 69.8%]
Prompt-3	61.2%	[54.6%, 67.5%]
Prompt-4	56.1%	[49.5%, 62.7%]

5 Results

In this section, we present the evaluation results of the proposed cybersecurity training framework, *CertiGuard*. Our analysis focuses on three primary aspects: the impact of various prompt designs, the effectiveness of advanced reasoning techniques, and the comparison of LoRA versus QLoRA in training efficiency.

5.1 Evaluation metrics and statistical uncertainty

Model performance is reported as passing rate/accuracy on certification-specific exam-style test sets. Because each test set is small (52–56 items), we add 95% Wilson confidence intervals to quantify uncertainty. For aggregate comparisons across the four certifications, we use the total number of evaluated items ($N = 215$). These intervals are intended to prevent overinterpretation of small differences. Because the study uses restricted third-party question text and does not redistribute item-level prediction vectors, the analysis emphasizes confidence intervals and conservative interpretation rather than making strong claims of statistical significance.

5.2 Prompt evaluation

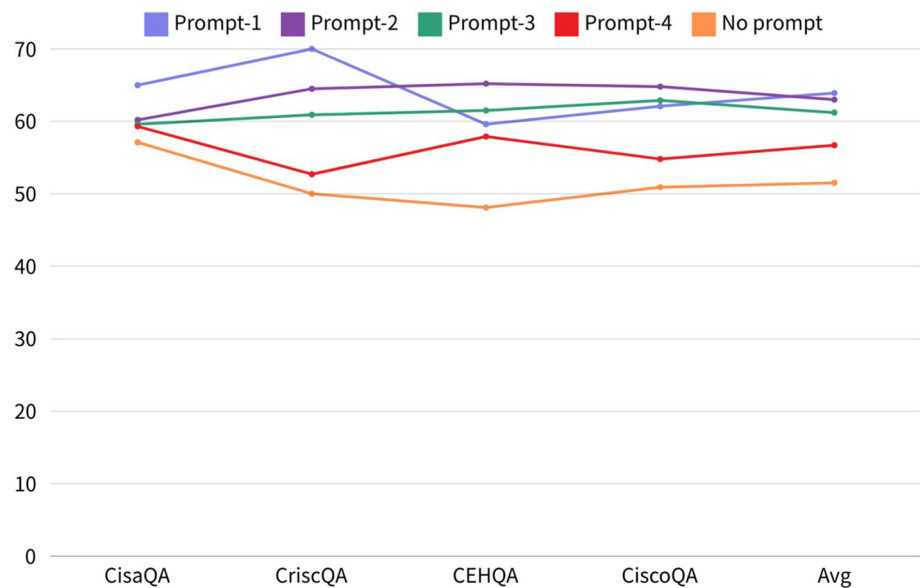
To investigate how prompt design influences model performance, we evaluated multiple prompt configurations on *Certibase*. As shown in Table 8, five different prompt settings (No Prompt, Prompt-1, Prompt-2, Prompt-3, and Prompt-4) were tested across four certification domains: CISAQA, CRISCQA, CEHQA, and CCNAQA. Figure 3 illustrates the passing rates achieved under these various prompt designs.

Observations:

- **Prompt-1** achieved the highest average passing rate (64.1%), demonstrating the importance of structured guidance.
- **Prompt-2** exhibited strong performance in CEHQA (65.2%), suggesting that detailed, specific instructions particularly benefit certain technical domains.

Table 8 Certibase performance comparison across different prompts

Prompts	CISAQA	CRISCQA	CEHQA	CCNAQA	Avg
No prompt	57.1%	50.0%	48.1%	50.9%	51.5%
Prompt-1	65.0%	70.0%	59.6%	62.1%	64.1%
Prompt-2	60.2%	64.5%	65.2%	64.8%	63.6%
Prompt-3	59.6%	60.9%	61.5%	62.9%	61.2%
Prompt-4	59.3%	52.7%	57.9%	54.8%	56.1%

Fig. 3 Certibase passing rate across different prompts

- The *No Prompt* condition yielded the lowest performance, underscoring the necessity of well-designed prompts to elicit accurate model responses.

Furthermore, an ablation study on subtle prompt variations indicates that even minor changes in phrasing and context can result in measurable performance differences. When comparing the best-performing prompt with the baseline, an improvement in the average passing rate was observed; however, the confidence intervals in Table 7 indicate that these gains should be interpreted cautiously because of the small test sets [17].

5.3 Advanced reasoning techniques

We further evaluated the impact of different Chain-of-Thought (CoT) strategies and advanced reasoning methods on model accuracy. Table 9 compares three models—LLaMA-2-7B, Certibase, and CertiGuard—across various CoT techniques, including standard CoT, Logic CoT, System 2 Attention (S2A), Contrastive CoT (CCoT), and Self-consistency CoT (SCoT).

Observations:

- *CertiGuard* achieved the highest average accuracy within each reasoning family, indicating that the combination of domain adaptation and QA-oriented supervised fine-tuning improves certification-style question answering under the current experimental setting.
- *Self-consistency CoT* produced the strongest aggregate result for CertiGuard (86.5%), followed by **Contrastive CoT** (84.7%) and **Logic CoT** (83.4%). However, the confidence intervals indicate that these differences should be interpreted cautiously.
- The results suggest different strengths across domains: logic-oriented reasoning is useful for governance, audit, and risk questions, whereas contrastive reasoning is helpful for technical domains such as CEH and CCNA, where distractor options often represent plausible but incorrect attack or networking scenarios.

From a deployment perspective, these results support a differentiated use of reasoning prompts. Logic CoT is appropriate when learners must validate control relationships, audit judgments, or risk responses. Contrastive CoT is useful when learners must compare similar technical alternatives, such as attack vectors, protocol behaviors, or configuration choices. SCoT can improve robustness by sampling multiple reasoning paths, but it increases inference cost and should therefore

Table 9 Reasoning performance across multiple CoT techniques with aggregate 95% Wilson confidence intervals

Model	CISAQA	CRISCQA	CEHQA	CCNAQA	Avg	Avg. 95% CI
<i>Chain-of-Thought</i>						
LLaMA-2-7B	70.7%	74.3%	81.4%	83.5%	77.4%	[71.4%, 82.5%]
Certibase	72.3%	74.9%	82.0%	84.7%	78.4%	[72.4%, 83.4%]
CertiGuard	76.7%	76.3%	84.9%	87.0%	81.1%	[75.3%, 85.8%]
<i>Logic Chain-of-Thought</i>						
LLaMA-2-7B	80.4%	76.8%	84.3%	75.0%	79.1%	[73.2%, 84.0%]
Certibase	81.5%	77.5%	85.1%	76.6%	80.1%	[74.3%, 84.9%]
CertiGuard	87.5%	80.0%	86.7%	79.6%	83.4%	[77.8%, 87.8%]
<i>System 2 Attention</i>						
LLaMA-2-7B	69.4%	70.1%	79.1%	74.6%	73.3%	[67.0%, 78.8%]
Certibase	71.4%	70.9%	80.7%	75.3%	74.5%	[68.3%, 79.9%]
CertiGuard	74.0%	72.7%	83.0%	77.7%	76.8%	[70.7%, 81.9%]
<i>Contrastive Chain-of-Thought</i>						
LLaMA-2-7B	71.5%	76.6%	85.4%	86.7%	80.0%	[74.1%, 84.8%]
Certibase	72.3%	77.2%	87.5%	88.1%	81.2%	[75.4%, 85.9%]
CertiGuard	75.9%	78.1%	92.4%	92.5%	84.7%	[79.3%, 88.9%]
<i>Self-consistency Chain-of-Thought</i>						
LLaMA-2-7B	79.8%	77.5%	84.7%	87.6%	82.4%	[76.8%, 86.9%]
Certibase	80.7%	79.6%	86.3%	88.5%	83.7%	[78.2%, 88.0%]
CertiGuard	83.3%	81.8%	90.5%	90.7%	86.5%	[81.3%, 90.4%]

be used selectively for difficult questions or formative feedback rather than every practice item.

5.4 Qualitative reasoning and error analysis

To contextualize the quantitative results, we examined representative model outputs from the reasoning experiments. Table 10 summarizes common qualitative patterns. Structured reasoning prompts were most useful when a question required distinguishing similar concepts, such as clickjacking versus cross-site request forgery, or risk acceptance versus risk mitigation. However, reasoning prompts did not eliminate errors: some outputs still selected a plausible but incorrect distractor, overemphasized irrelevant scenario details, or produced confident explanations for wrong answers.

As a sanitized example, a scenario involving a logged-in banking session, a malicious link, and an unauthorized background transaction requires the model to identify whether the core mechanism is clickjacking, cross-site scripting, or cross-site request forgery. A standard answer-only prompt sometimes focused on the visible link or video content, whereas a contrastive reasoning prompt more explicitly compared why clickjacking was insufficient and why the unauthorized request pattern better matched CSRF. This analysis helps explain why contrastive and logic-oriented reasoning prompts can improve performance, while also

Table 10 Qualitative error patterns observed in reasoning outputs

Error pattern	Description
Concept confusion	The model confused related security concepts, especially when options shared overlapping terminology
Distractor attraction	The model selected an option that matched a keyword in the question but did not address the core scenario
Incomplete causal reasoning	The model identified part of the attack chain but failed to connect the sequence of events to the correct vulnerability
Overconfident explanation	The model generated a fluent explanation despite selecting the wrong option
Domain imbalance	Performance varied across managerial and technical domains, suggesting that certification-specific coverage still matters

showing that reasoning text should be interpreted as an explanatory aid rather than a guarantee of correctness.

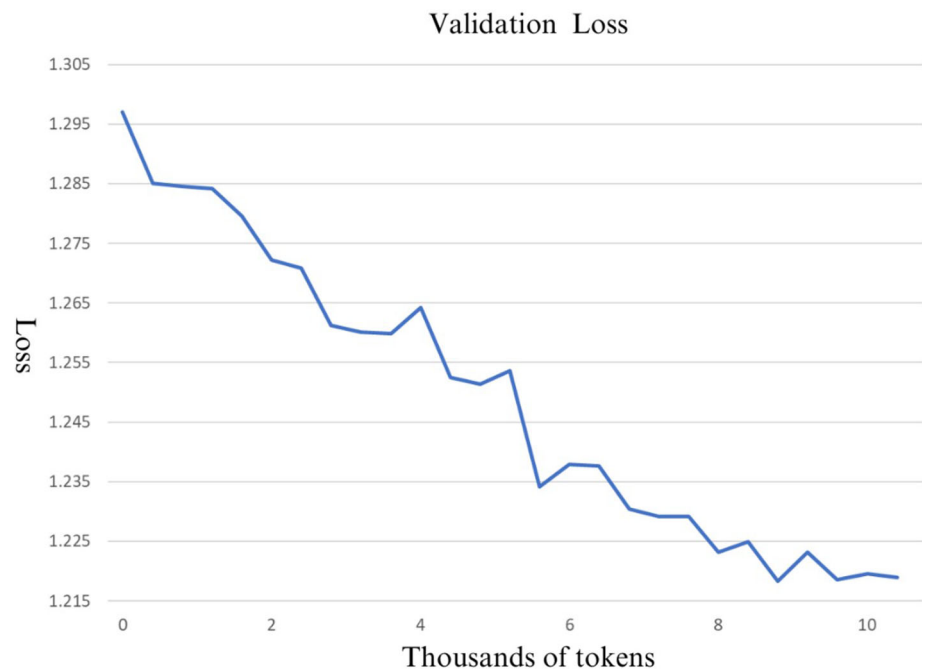
5.5 LoRA vs. QLoRA training efficiency

To assess training efficiency, we compared the performance of LoRA and QLoRA in terms of training and validation loss over multiple epochs. Figure 4 illustrates the training loss curves, while Fig. 5 presents the corresponding validation loss curves.

Fig. 4 Comparison of training loss: LoRA vs. QLoRA



Fig. 5 Validation loss over training epochs



Observations:

- *QLoRA* exhibits a steeper decline in training loss, indicating faster convergence and enhanced memory efficiency compared to LoRA [8].
- The validation loss curves confirm that both methods generalize well; however, QLoRA maintains a slight advantage in consistency over extended training.
- The significant reduction in VRAM usage with QLoRA not only facilitates scaling to larger models but also enables more complex tasks to be trained without substantial hardware cost increases.

5.6 Comprehensive analysis

The overall experimental results underscore the potential of our domain-adaptive training framework for cybersecurity applications. Key findings include:

- Structured prompt designs (e.g., Prompt-1, Prompt-2) yield measurable improvements over unstructured prompts, although the small test sets require cautious interpretation [17].
- Advanced reasoning strategies, particularly those incorporating logical and contrastive elements, enhance the

model's capability to perform multi-step reasoning in complex scenarios [18–20].

- The QLoRA training approach offers superior efficiency and scalability, as evidenced by faster convergence and reduced memory consumption [8].

Collectively, these findings demonstrate that integrating advanced large language model techniques—from prompt engineering to efficient training algorithms—can effectively bridge the cybersecurity skills gap by providing scalable, cost-effective, and personalized training solutions.

6 Discussion

This section discusses the implications of our findings, evaluates the strengths and limitations of the proposed approach, and outlines directions for future research in applying LLMs to cybersecurity certification training.

6.1 Summary of findings

The results suggest that domain-specific fine-tuning, structured prompts, and reasoning-oriented prompting can improve performance on cybersecurity exam-style questions. QLoRA also makes the training process more feasible under limited computational resources. However, these improvements should be treated as exploratory because the evaluation sets are small, third-party question text cannot be redistributed, and direct human or commercial-tool baselines are not available under a common benchmark.

6.2 Explainability and trustworthiness

CertiGuard provides explanations because explanations are useful for training and formative feedback. Nevertheless, model-generated reasoning is not equivalent to a verified proof. Chain-of-Thought outputs may contain plausible but incorrect steps, and a correct answer may sometimes be accompanied by incomplete reasoning. For this reason, CertiGuard should be deployed with answer keys, instructor review, or retrieval-grounded references when used in formal training. Trustworthiness can be improved through confidence calibration, answer verification, retrieval-augmented grounding, and logging of model rationales for audit and error analysis.

6.3 Bias and fairness considerations

Bias may appear across certification domains, learner levels, and question styles. A model trained on more technical questions may underperform on governance or risk-management questions, while a model trained on managerial content may

struggle with packet-level networking or exploitation scenarios. Future evaluations should report performance by domain, difficulty, and question type, rather than only average passing rates. Fairness also requires that learners are not misled into believing that model feedback is equivalent to official certification training or guaranteed exam success.

6.4 Security, privacy, and ethical considerations

LLM-based education systems introduce several security and ethical risks. Prompt injection may cause the model to ignore learning instructions or reveal restricted content. Data poisoning may influence the training corpus if untrusted materials are included. Memorization may lead to the reproduction of copyrighted or confidential exam-style text. Model misuse is also possible if offensive security explanations are presented without appropriate safeguards. To mitigate these risks, CertiGuard should use curated sources, remove restricted content from public releases, filter harmful outputs, avoid redistributing third-party exam items, and provide clear usage policies. These safeguards are especially important in cybersecurity education, where dual-use knowledge can support both defensive learning and misuse.

6.5 Practical case study: academic and professional training

A practical deployment scenario is a university or corporate cybersecurity training course. In such a setting, CertiGuard can be used as a supplementary practice assistant. Beginner learners may ask for definitions and simplified explanations, intermediate learners may work through scenario-based questions, and advanced learners may request step-by-step comparisons among similar attack or control concepts. Instructors can use the system to generate draft practice questions, explanations, and remediation hints, but final materials should be reviewed by human experts. The system's value is therefore best understood as improving training scalability and feedback availability, not as replacing certified instructors or official exam preparation products.

6.6 Limitations and challenges

This study has several limitations. First, the test sets are small, which limits statistical power and generalizability. Second, the evaluation data are based on certification-style materials that cannot be fully redistributed because of copyright and exam-security restrictions. Third, the possibility of memorization cannot be completely eliminated without larger public benchmarks and item-level audits. Fourth, the proposed proficiency-tiered deployment is static and does not yet implement continuous learner modeling or adaptive recommendation. Fifth, no direct comparison with

human examinees or commercial certification tools is provided, because there is no shared public benchmark for such comparisons and commercial systems are closed. Finally, the model's performance may degrade when certification domains change or when new cybersecurity topics emerge.

6.7 Future directions

Future work will focus on three directions. First, we plan to construct a legally redistributable evaluation set based on certification blueprints and expert-authored exam-style questions, which would enable stronger reproducibility and more rigorous statistical testing. Second, future versions of CertiGuard should incorporate retrieval-augmented generation, answer verification, and calibration mechanisms to improve trustworthiness. Third, the proficiency-tiered strategy should be extended into a truly adaptive learning system that tracks learner performance over time, identifies weak knowledge areas, and recommends personalized remediation activities. Expansion to other certification domains, such as cloud security, secure software development, and incident response, is also a promising direction.

7 Conclusion

This paper presented CertiGuard, a QLoRA-based cybersecurity training framework that adapts LLaMA-2-7B to certification-oriented learning across CISA, CRISC, CEH, and CCNA domains. The framework combines continued domain adaptation, supervised fine-tuning, certification-aligned instructions, proficiency-tiered deployment, and reasoning-oriented prompt strategies.

The results indicate that structured prompts and domain-specific fine-tuning can improve exam-style question answering; in the reasoning comparison, CertiGuard achieved its highest aggregate performance with Self-consistency CoT (86.5%). Reasoning prompts can also provide more informative explanations for learners. At the same time, the findings should be interpreted conservatively because the evaluation sets are small and the underlying third-party question text cannot be redistributed. To improve transparency, this study clarifies data provenance and copyright constraints, reports statistical uncertainty, provides prompt templates, and discusses limitations, ethics, and future work.

Overall, CertiGuard represents a step toward scalable AI-supported cybersecurity education. Its most appropriate role is as a supplementary training assistant that helps learners practice concepts, compare reasoning paths, and receive explanatory feedback under human or institutional oversight. Future work will prioritize legally redistributable benchmarks, stronger statistical evaluation, adaptive learner modeling, and safer deployment mechanisms.

Appendix A Prompt and Reasoning Templates

This appendix provides the prompt and reasoning templates used in this study. The example question text below is a sanitized illustrative item written for transparency and should not be interpreted as an official certification question.

A.1 Prompt Design Templates

- **No Prompt:** Provide the question and answer options only. The model selects the most likely correct answer.
- **Prompt-1: Structure and Clarity:** Provide clear task instructions, state the expected answer format, and ask the model to identify the best option based on the scenario.
- **Prompt-2: Specificity and Information:** Provide the scenario, ask for the most accurate answer, and request a concise explanation of why the answer is correct.
- **Prompt-3: User Interaction and Engagement:** Present the question in a learner-facing style and ask the model to explain the selected answer in an instructional manner.
- **Prompt-4: Content and Language Style:** Present the question using formal, concise language and ask the model to choose the best option.

A.2 Reasoning Templates

- **Chain-of-Thought (CoT):** Ask the model to consider the scenario step by step before selecting the answer.
- **Logic CoT:** Ask the model to verify each causal step and check whether the selected option logically follows from the evidence in the scenario.
- **System 2 Attention (S2A):** Ask the model to focus only on the key actions and ignore distracting details.
- **Contrastive CoT:** Ask the model to compare a plausible correct reasoning path with at least one incorrect reasoning path before selecting the final answer.
- **Self-consistency CoT:** Ask the model to generate multiple reasoning paths and select the most consistent final answer.

Acknowledgements We acknowledge the open-source communities of Hugging Face Transformers for providing essential toolkits that enabled efficient model fine-tuning. This work was supported in part by access to Google Colab computational resources.

Funding This research was supported by the National Science and Technology Council (NSTC), Taiwan (Grant Nos. 113-2221-E-029-009-MY3 and 114-2634-F-027-001-MBK). The funder had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not applicable.

Consent for publication Not applicable.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., et al.: Attention is all you need. *Adv. Neural. Inf. Process. Syst.* **30**, 5998–6008 (2017)
2. Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., et al.: Language models are few-shot learners. *Adv. Neural. Inf. Process. Syst.* **33**, 1877–1901 (2020)
3. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., et al.: Training language models to follow instructions with human feedback. *Adv. Neural. Inf. Process. Syst.* **35**, 27730–27744 (2022)
4. Kasneci, E., Sessler, K., Kuechemann, S., Bannert, M., Dementieva, D., Fischer, F., et al.: ChatGPT for good? On opportunities and challenges of large language models for education. *Learn. Individ. Differ.* **103**, 102274 (2023). <https://doi.org/10.1016/j.lindif.2023.102274>
5. Al Shloul, T., Mazhar, T., Abbas, Q., Iqbal, M., Ghadi, Y.Y., Shahzad, T., et al.: Role of activity-based learning and ChatGPT on students' performance in education. *Comput. Educ.: Artif. Intell.* **6**, 100219 (2024). <https://doi.org/10.1016/j.caeai.2024.100219>
6. Shahzad, T., Mazhar, T., Tariq, M.U., Ahmad, W., Ouahada, K., Hamam, H.: A comprehensive review of large language models: issues and solutions in learning environments. *Discover Sustainab.* **6**(1), 27 (2025). <https://doi.org/10.1007/s43621-025-00815-8>
7. Khan, S., Mazhar, T., Shahzad, T., Khan, M.A., Rehman, A.U., Saeed, M.M., et al.: Harnessing AI for sustainable higher education: ethical considerations, operational efficiency, and future directions. *Discover Sustain.* **6**(1), 23 (2025). <https://doi.org/10.1007/s43621-025-00809-6>
8. Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L.: Qlora: Efficient finetuning of quantized llms. *Adv. Neural. Inf. Process. Syst.* **36**, 10088–10115 (2023)
9. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., et al. (2022) Lora: Low-rank adaptation of large language models. *ICLR*. 1 (2): 3. <https://doi.org/10.48550/arXiv.2106.09685>
10. Guo, Z., Hua, Y.: Continuous training and fine-tuning for domain-specific language models in medical question answering. *arXiv preprint arXiv:2311.00204* (2023). <https://doi.org/10.48550/arXiv.2311.00204>
11. Nachane, S.S., Gramopadhye, O., Chanda, P., Ramakrishnan, G., Jadhav, K.S., Nandwani, Y., et al.: Few shot chain-of-thought driven reasoning to prompt LLMs for open ended medical question answering. *arXiv preprint arXiv:2403.04890* (2024). <https://doi.org/10.18653/v1/2024.findings-emnlp.31>
12. Chen, Z.Y., Xie, F.K., Wan, M., Yuan, Y., Liu, M., Wang, Z.G., et al.: MatChat: A large language model and application service platform for materials science. *Chin. Phys. B* **32**(11), 118104 (2023). <https://doi.org/10.1088/1674-1056/ad04cb>
13. Abbas Shah, S.F., Mazhar, T., Shahzad, T., Khan, M.A., Ghadi, Y.Y., Hamam, H.: Integrating educational theories with virtual reality: Enhancing engineering education and VR laboratories. *Social Sci. Humanit. Open.* **10**, 101207 (2024). <https://doi.org/10.1016/j.ssaho.2024.101207>
14. Aghaei, E., Niu, X., Shadid, W., Al-Shaer, E.: Securebert: A domain-specific language model for cybersecurity. In: *International Conference on Security and Privacy in Communication Systems*. Springer; (2022). p. 39–56
15. Jin, J., Tang, B., Ma, M., Liu, X., Wang, Y., Lai, Q., et al.: Crimson: Empowering strategic reasoning in cybersecurity through large language models. *arXiv preprint arXiv:2403.00878*. (2024) <https://doi.org/10.48550/arXiv.2403.00878>
16. Tann, W., Liu, Y., Sim, J.H., Seah, C.M., Chang, E.C.: Using large language models for cybersecurity capture-the-flag challenges and certification questions. *arXiv preprint arXiv:2308.10443*. (2023) <https://doi.org/10.48550/arXiv.2308.10443>
17. Bsharat, S.M., Myrzakhan, A., Shen, Z.: Principled instructions are all you need for questioning llama-1/2, gpt-3.5/4. *arXiv preprint arXiv:2312.16171*. (2023) 3. <https://doi.org/10.48550/arXiv.2312.16171>
18. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., et al.: Chain-of-thought prompting elicits reasoning in large language models. *Adv. Neural. Inf. Process. Syst.* **35**, 24824–24837 (2022)
19. Chia, Y.K., Chen, G., Tuan, L.A., Poria, S., Bing, L.: Contrastive chain-of-thought prompting. *arXiv preprint arXiv:2311.09277*. (2023) <https://doi.org/10.48550/arXiv.2311.09277>
20. Zhao, X., Li, M., Lu, W., Weber, C., Lee, J.H., Chu, K., et al.: Enhancing zero-shot chain-of-thought reasoning in large language models through logic. *arXiv preprint arXiv:2309.13339*. (2023) <https://doi.org/10.48550/arXiv.2309.13339>
21. Weston, J., Sukhbaatar, S.: System 2 attention (is something you might need too). *arXiv preprint arXiv:2311.11829*. (2023) <https://doi.org/10.48550/arXiv.2311.11829>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.